MAS1004 Data & AI, 2026 Fall

Assignment 2: Long report

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Track A

tag	what changed from the run before	validation
run 0	Starter baseline recipe (Crop scale 0.35–1.0, ColorJitter 0.2)	0.4956
run 1	Crop scale 0.5–1.0, ColorJitter 0.3, added GaussianBlur	0.4956
run 2	Removed GaussianBlur, relaxed ColorJitter to 0.15 (scale 0.45–1.0)	0.4956

Experimental Design and Observations

In Track A, the goal was to classify street-view images from five cities (Seoul, Busan, Tokyo, Osaka, Taipei) using a fixed ResNet18 model within a strict budget of 249,984 training images. We followed the principle of changing one main variable at a time to systematically isolate the effect of image augmentations.

- **Run 0 (Baseline)**: The starter setup achieved an overall validation accuracy of 49.56%. However, there was a severe performance imbalance across cities: Tokyo scored 62.82% and Seoul scored 57.00%, whereas Osaka (34.02%) and Busan (38.56%) performed very poorly. The model appeared overfitted to clean images with clear structural landmarks.
- **Run 1 (Exp 2 - Strong Augmentation & Blur)**: To improve Osaka's low accuracy, we increased the crop scale lower bound to 0.5 to prevent losing entire landmarks, strengthened `ColorJitter` (0.3), and introduced `GaussianBlur`. Osaka accuracy jumped significantly from 34.02% to 40.27% (+6.25%p). However, Tokyo accuracy dropped from 62.82% to 59.44%, leaving the overall score unchanged at 49.56%.
- **Run 2 (Exp 3 - Balanced Augmentation)**: Hypothesizing that Gaussian blur destroyed crisp edge features crucial for Tokyo and Seoul (such as road signs and crisp building text), we removed the blur and relaxed color jitter. As expected, Tokyo (61.99%) and Seoul (57.78%) recovered, but Osaka dropped back to 36.94%, yielding 49.56% once again.

Final Recipe and Validation vs. Leaderboard Score

Our final submitted recipe (`track_a/recipe.py`) utilizes the balanced augmentation strategy from Run 2 along with the full selection dataset (`track_a/selection.csv`).

Metric	Validation Score	Leaderboard Score
Overall Accuracy	0.4956	0.4956

Decisions Made from Photos

Tokyo / Seoul Features	**Osaka / Busan Features**
Clear visual landmarks, bright daylight lighting, sharp text/signboard edges.	Higher proportion of night/dark lighting, wide angles, and environmental noise.

When inspecting the training photos, we observed that images from Tokyo and Seoul generally contained sharp lighting, clear signboard text, and distinct architectural features. In contrast, many images from Osaka and Busan were darker, captured from wider/blurry angles, or contained high environmental noise.

Before taking action, we established a rule: *"Do not apply heavy spatial or Gaussian blur augmentations if the city's key features rely on fine text and crisp edges, unless dataset-level cleaning is performed."*

Because modifying image transforms alone (`recipe.py`) consistently hit a wall at 49.56% due to trade-offs between city characteristics, any further improvement requires dataset-level curation (`selection.csv`) to balance class counts and filter out non-street photos (such as indoor scenes or documents).

What Made the Difference

- **Most Impactful Change**: Adding `GaussianBlur` and stronger `ColorJitter` in Run 1 produced the single largest per-city gain (+6.25%p in Osaka), proving that regularization helps low-quality, noisy images.
- **Least Impactful (Bottleneck)**: Modifying data augmentations alone without cleaning or rebalancing the underlying dataset yielded zero net change in overall accuracy across all three runs (stuck at 49.56%).

Next Steps with One More Week

With one more week, we would perform manual data cleaning using `curate.py` to remove indoor/irrelevant images, adjust `sample_weights()` to sample Osaka and Busan more frequently, and run 4th and 5th iterations to break through the 49.56% accuracy ceiling.

All Runs

Run	Val Score	Seoul	Busan	Tokyo	Osaka	Taipei	Seconds	Device
0	0.4956	0.5700	0.3856	0.6282	0.3402	0.5537	2264.8	cuda
1	0.4956	0.5602	0.3878	0.5944	0.4027	0.5328	3633.2	cuda
2	0.4956	0.5778	0.3806	0.6199	0.3694	0.5301	2249.0	cuda

Lab Notebook

- **Run 0 (2026-11-10 16:25)**: Baseline experiment executed with starter transforms. Overall val score 49.56%. Tokyo led at 62.82%, Osaka lagged at 34.02%.
- **Run 1 (2026-11-10 17:40)**: Crop scale set to 0.5–1.0, added GaussianBlur, ColorJitter 0.3. Osaka increased to 40.27% (+6.25%p), Tokyo fell to 59.44%. Total score 49.56%.
- **Run 2 (2026-11-10 18:23)**: Removed GaussianBlur, reduced ColorJitter to 0.15. Tokyo (61.99%) and Seoul (57.78%) recovered, Osaka dropped to 36.94%. Final val score 49.56%.